H

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY

9648/03

Paper 3 Applications and Planning

26 September 2016

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.

For Examiner's Use	
1	
2	
3	
5	
Total	60
For Examiner's Use	
4	12

This document consists of **12** printed pages and **0** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

1 Insulin deficiency is one of the causes of diabetes. There are a few methods to synthesize recombinant insulin using genetic engineering, one of the methods involve cloning the human insulin cDNA.

(a) Compare the differences on how a genomic library and a cDNA library are produced.

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[2]

(b) Explain the advantages of obtaining human insulin gene from a cDNA library instead of a genomic DNA library.

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[3]

Fig. 1.1 shows the bacteria plasmid containing the positions of the restriction sites of the restriction enzymes available at its multiple cloning site. These restriction sites do not occur elsewhere within the plasmid.

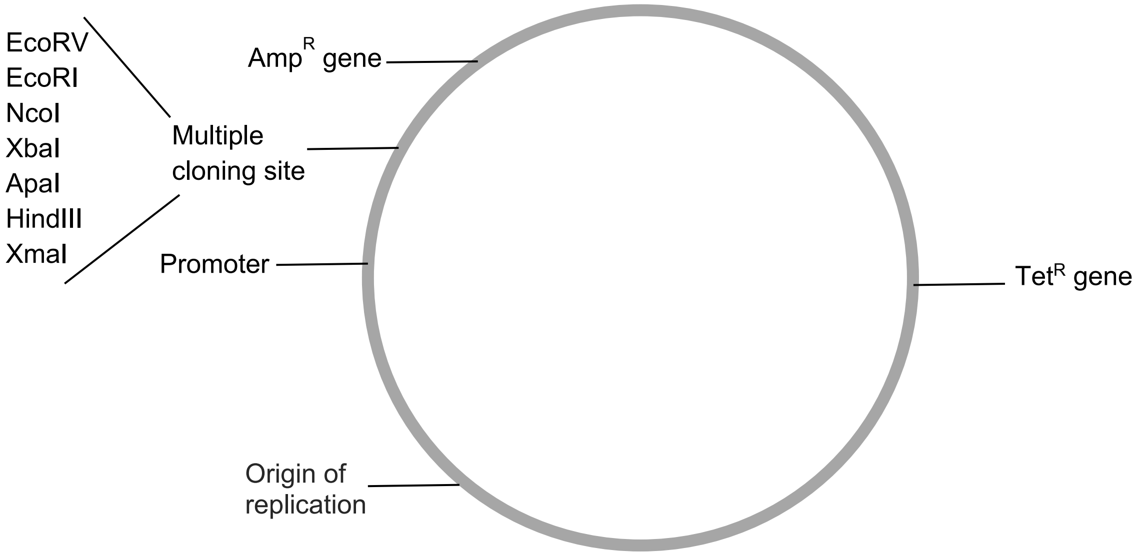

Fig. 1.1

Fig. 1.2 shows the restriction enzymes and their corresponding specific recognition sequences with respect to human insulin cDNA sequence. With the exception of *EcoRV*, all the restriction sites (*Apa I*, *Nco I*) generate sticky ends when cut.

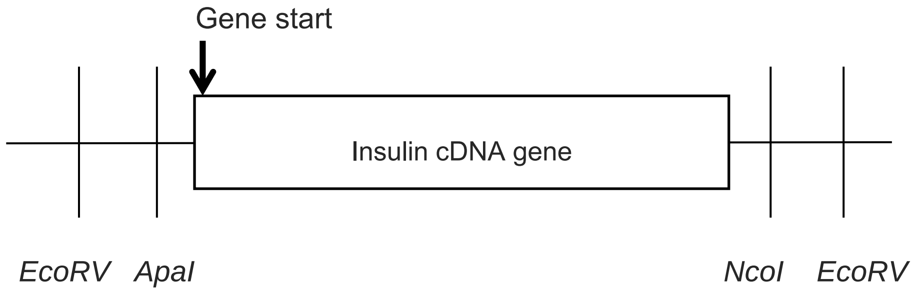

Fig. 1.2

(c) In order to obtain a recombinant plasmid containing the human insulin gene, the human insulin cDNA and vector needs to be digested with restriction enzymes prior to annealing them together.

(i) State the restriction enzymes used in order to obtain **efficient** gene expression.

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[1]

(ii) Explain your choice of restriction enzymes.

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.... [2]

(iii) With reference to **Fig. 1.1** and **1.2**, explain how recombinant bacteria (formed using the recombinant plasmids) can be selected using antibiotics.

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.... [4]

(d) Suggest a reason why functional insulin was not expressed in the recombinant E.coli cells.

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.... [2]

(e) Bacteriophages can be used to transfer foreign DNA into bacteria cells. Suggest **one** advantage of the use of phage vectors over plasmid vectors in transferring DNA into bacteria.

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.... [1]

[Total: 15]

2 A violent air crash resulted in the death of 150 passengers. The poor condition of the crash victims made physical identification impossible. Forensic scientists were tasked to confirm the identities of victims by using molecular techniques for families who have come forward to claim the correct remains of their relatives.

DNA samples of human remains and surviving relatives were subjected to restriction enzyme digest and Southern blot analysis to analyse VNTR band patterns. Single locus VNTR probes will yield DNA patterns of seven individuals at a single locus, as shown in **Fig. 2.1**. On the other hand, multi-locus VNTR probes can yield more information about each individual since 10 to 30 loci can be simultaneously analysed. This is illustrated in five different individuals in **Fig. 2.2**.

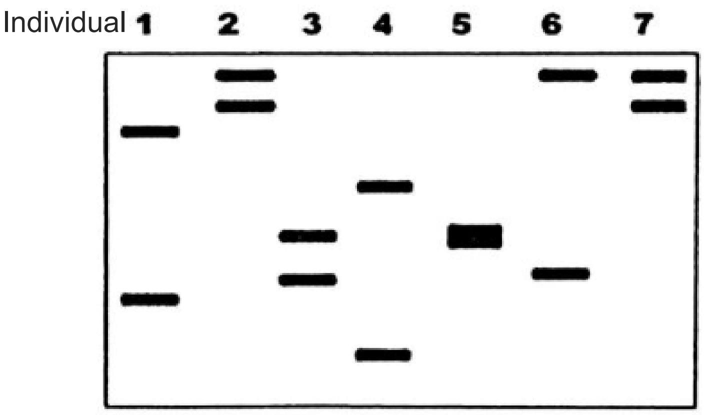

Fig. 2.1

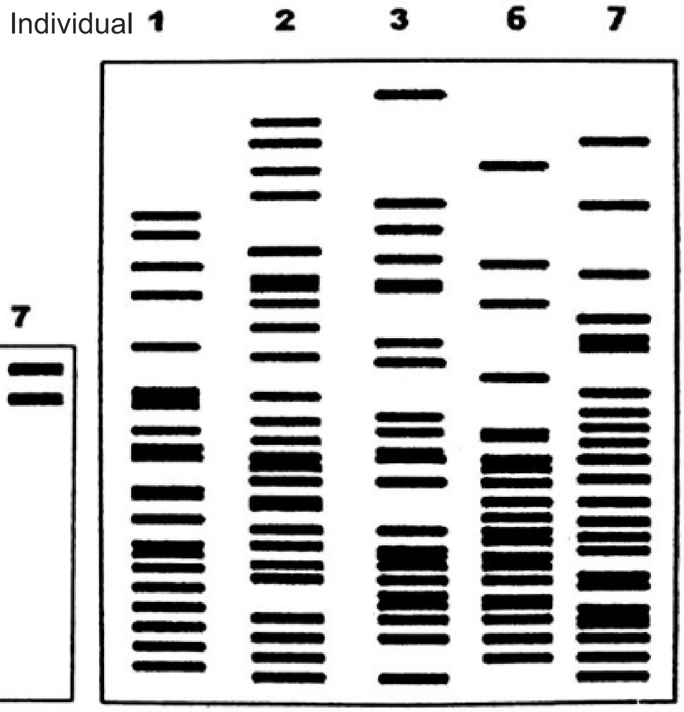

Fig. 2.2

(a) **(i)** Explain the basis of application of VNTR in DNA profiling.

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(ii) With reference to **Fig. 2.1**, explain the different numbers of fragments seen in different individuals.

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(iii) With reference to **Fig. 2.1** and **Fig. 2.2**, suggest why using multi-locus probes is preferred.

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.... [2]

(b) VNTRs can be analysed by Southern blotting or by PCR analysis.

Describe **two** differences between Southern blotting and PCR analysis in the analysis of the single locus VNTR shown in **Fig. 2.1.**

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(c) The fully sequenced human genome contains many unknown genes. Suggest how RFLP analysis can help to identify **unknown** genes that are responsible for heritable diseases.

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[Total: 12]

3 Plant tissue culture is a method used to propagate plants. **Fig. 3.1** shows one method of plant tissue culture.

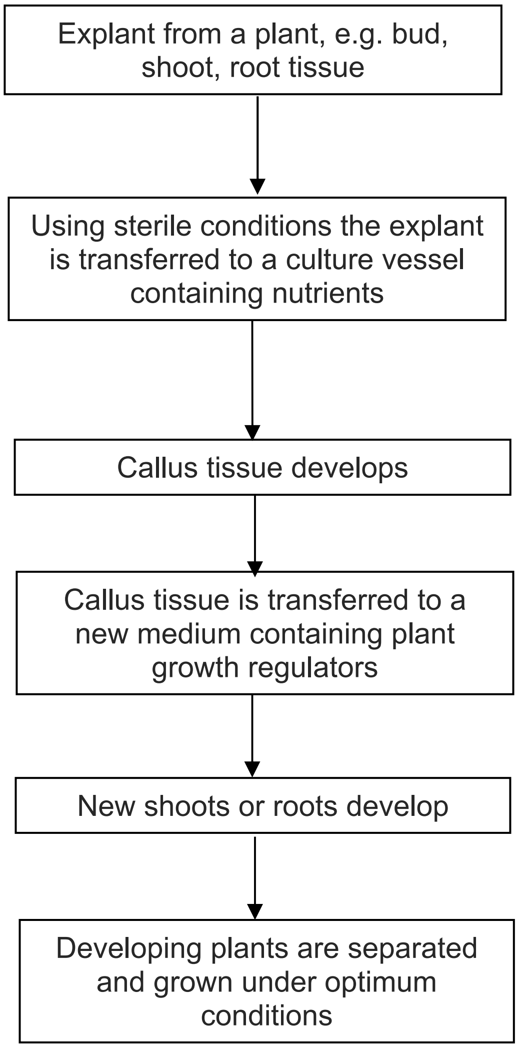

Fig. 3.1

(a) Suggest why explants are used in tissue culture.

[1]

(b) Suggest why the explant is initially grown in sterile conditions.

[2]

After the callus tissue develops, equal masses of the plant callus were cultured for four weeks on media containing different concentrations of two plant growth regulators: auxin and cytokinin. The results are shown in **Fig. 3.2.**

Treatment	Concentration of plant growth regulators/ mg dm^{-3}		Effect of plant growth regulators on callus growth
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

Fig. 3.2

(c) Explain why growing plantlets from a callus in tissue culture results in a clone.

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[3]

(d) With reference to **Fig. 3.2**, describe the effects of auxin and cytokinin on callus growth.

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[3]

(e) Outline using **one** named example, the process of genetic engineering that could increase crop yield.

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.... [4]

(f) Suggest one possible risks to the environment of growing genetically engineered crops.

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[Total: 14]

Planning Question

4 Amylases are naturally found in wheat flour. During bread-making when water is added to flour, amylases are activated and break down starch in flour into maltose. Maltose is a reducing sugar and its presence can be tested with Benedict's solution.

Amylase activity can be inhibited by heavy metals ions such as iron (Fe^{2+}). Using this information and your knowledge, design an experiment to determine if Fe^{2+} functions as a competitive inhibitor or a non-competitive inhibitor and its effect on the rate of amylase activity.

You must use:

- 1.0% stock α-amylase solution
- 0.3% iron sulfate solution
- 5.0% starch solution
- distilled water
- Benedict's solution
- 10.0% maltose solution

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc. ,
- syringes,
- white card,
- white tile,
- blunt forceps,
- Bunsen burner with tripod, gauze and bench mat,
- thermometer.
- water bath
- timer e.g. stopwatch or stop clock

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Explain how a disease such as cystic fibrosis can be treated by gene therapy, using **non-viral** delivery systems. [6]

(b) Explain, with specific examples, how genetic engineering can improve the quality and yield of crop plants and animals in solving the demand for food in the world. [7]

(c) Discuss the ethical and social implications of genetically modified organisms. [7]

[Total: 20]